

Direct On Line (DOL) Motor MAPS Template (Electrical)

The Direct On Line (DOL) Motor MAPS template is designed to control a motor that is connected as a DOL.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the DOL motor.

This document describes the Advanced DOL Motor MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the DOL motor being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

This document also provides a typical electrical schematic for the Advanced DOL Motor MAPS Template.

Advanced DOL Motor Functional Philosophy

The DOL start output is enabled once all interlocks are ready and a start request is received.

Field Interlocks

- MCC (Overload) Trip Feedback
- Isolator On Feedback
- Safety Interlock Feedback
- Motor Current Feedback (Analog Input) (Current above High Limit Setpoint)

PLC Interlocks

- Start Interlock Ready
- Process Interlock Ready
- Restart Delay Interlock
- There are no interlock faults latched

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing on the reset button will unlatch the interlock and display a grey indication and allow the DOL to become ready for operation.

Maintenance mode - will disable the Start and Process Interlocks, but respects all the other interlocks. This is used to test a DOL while the process is stopped.

Simulation mode - will disable the physical output of the DOL and simulate an input run feedback after the elapsed setpoint time from the faceplate. The Start and Process Interlocks are respected. Simulation mode is used to test the process logic without starting the physical device itself. It is very useful in testing, commissioning and maintenance.

Start and stop requests are initiated by selecting a mode on the DOL faceplate:

- **Auto mode** - the auto control program initiates a start and stop, normal PLC control.
- **Manual mode & Desk mode** - the operator can start and stop from the faceplate.
- **Manual mode & Field mode** - the motor can be started and stopped from the start and stop push buttons in the field.

Typical Faceplate Graphics for the Advanced DOL Motor MAPS Template

DOL ISO	DOL MOTOR LEFT	DOL MOTOR RIGHT	DOL PUMP LEFT	DOL PUMP RIGHT
0000_ABCDE_123 I H M M	0000_ABCDE_123 H M S 	0000_ABCDE_123 H M S 	0000_ABCDE_123 H M S 	0000_ABCDE_123 H M S 
H	Manual Mode			
M	Maintenance Mode			
S	Simulation Mode			
I	Process Interlock			
0000_ABCDE_123	Motor Control Name			
Graphics legend:				
Line colour	Fill colour	DOL status		
Black	Grey	Stopped		
Grey	White	Stopping		
Green	White	Starting		
Black	Green	Running		
Black	Yellow/Red	Fault Active		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	x
SSCL 1	Fail To Start Setpoint	Analog	x	x	x
SSCL 2	Fail To Stop Setpoint	Analog	x	x	
SSCL 3	Running Setpoint	Analog	x	x	x
SSCL 4	Stopping Setpoint	Analog	x	x	
SSCL 5	Simulate Setpoint	Analog	x	x	x
SSCL 6	Restart Setpoint	Analog	x	x	
SSCL 7	Max Current Setpoint (High Limit)	Analog	x		
SSCL 8	Min In Current Setpoint (Raw Min)	Analog	x		
SSCL 9	Max In Current Setpoint (Raw Max)	Analog	x		
SSCL 10	Min Out Current Setpoint (Eng. Min)	Analog	x		
SSCL 11	Max Out Current Setpoint (Eng. Max)	Analog	x		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	x
SSSL 1	Starting Time Process Value (actual)	Analog	x	x	x
SSSL 2	Stopping Time Process Value (actual)	Analog	x	x	
SSSL 3	No of Operations (actual)	Analog	x	x	
SSSL 4	Restart Delay Time Remaining (actual)	Analog	x	x	
SSSL 5	Motor Current Process Value (actual)	Analog	x		
SSSL 6	Total Running Hours	Analog	x		

Digital Inputs:		Advanced	Standard	Basic
DI 0	Running Signal (On = Healthy)	x	x	x
DI 1	Isolator Healthy (On = Healthy)	x	x	
DI 2	MCC Healthy (On = Healthy)	x	x	x
DI 3	Safety Interlock Healthy (On = Healthy)	x	x	x
DI 4	Manual Start Push Button (On = Start)	x	x	
DI 5	Manual Stop Push Button (Off = Stop)	x	x	
Digital Outputs:				
DO 0	Device Start	x	x	x
Analog Input:				
AI 0	Analog Input for Current Drawn	x		

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Auto/Manual	x	x	x	x
Bit 1	Field/Desk	x	x	x	x
Bit 2	Maintenance	x	x	x	x
Bit 3	Simulation	x	x	x	x
Bit 4	Alarms All Reset		x	x	x
Bit 5	Reset No Operations Accumulator	x	x	x	
Bit 6	Start Command	x	x	x	x
Bit 7	Reset Running Hours Accumulator	x	x		
Bit 8	Stop Command	x	x	x	x
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Process Interlock		x	x	x
Bit 1	Ready To Start		x	x	x
Bit 2	Stopped		x	x	x
Bit 3	Starting		x	x	x
Bit 4	Running		x	x	x
Bit 5	Stopping		x	x	x
Bit 6	Fault		x	x	x
Bit 7	Start Interlock OK		x	x	x
Bit 8	Isolator Closed Fault	x	x	x	
Bit 9	Fault Fail To Stop	x	x	x	
Bit 10	Fault Fail To Start	x	x	x	x
Bit 11	MCC Fault	x	x	x	x
Bit 12	Safety Fault	x	x	x	x
Bit 13	Overcurrent Fault	x	x		

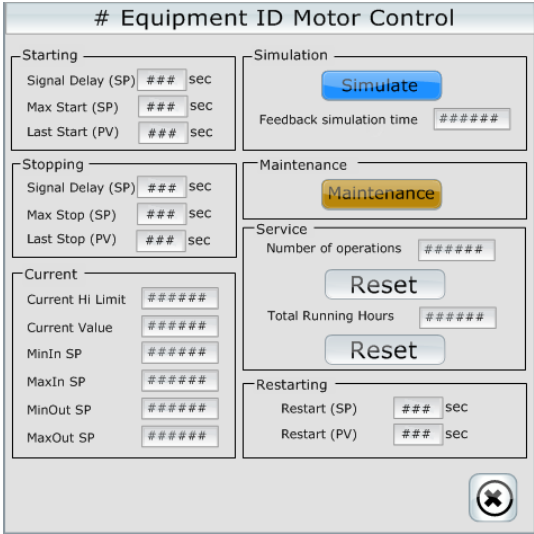
Bit 14	Reset Flash on Fault		x		
Bit 15	Fail To Start due to Interlock Alarm	x	x		

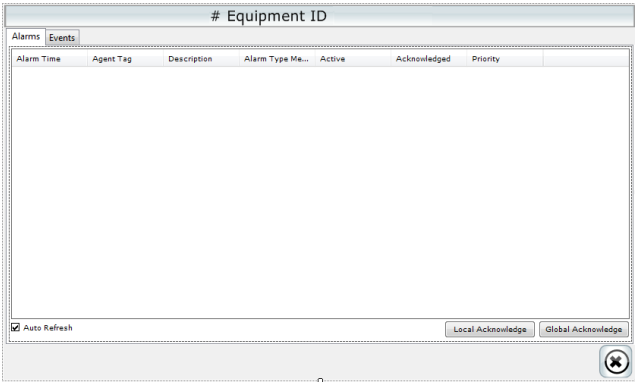
Resources	Advanced	Standard	Basic
Digital Inputs	6	6	3
Digital Outputs	1	1	1
Analog Inputs	1	0	0
Analog Outputs	0	0	0
PLC Memory Steps	518	299	205
System Words Used	53	22	19
System Bits Used	92	80	73
SCADA SCAN Points	19	12	6

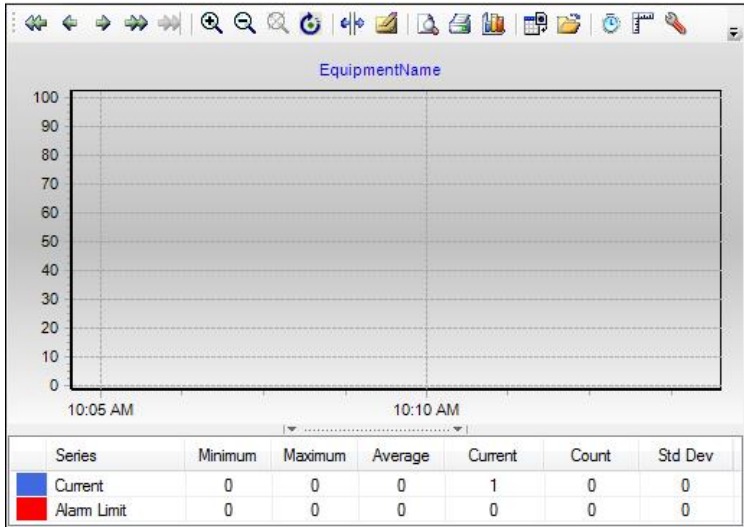
Template: Advanced DOL PLC Function Block

<Device> (Advanced Direct On Line Control)			
DATA_DOL_A_V1_0.F.I_Run_FB	F_I_Run_FB	FB:DOL_A_V1_0 DOL_A_v1_0	F_O_DOL_START_CMD
DATA_DOL_A_V1_0.F.I_Isolator_Closed_FB	F_I_Isolator_Closed_FB		S_SW1_Marshal
DATA_DOL_A_V1_0.F.I_MCC_Healthy_FB	F_I_MCC_Healthy_FB		S_SW2_Starting_Time_PV
DATA_DOL_A_V1_0.F.I_Safety_Interlock_OK	F_I_Safety_Interlock_OK		S_SW3_Stopping_Time_PV
DATA_DOL_A_V1_0.VAR_Process_Int_OK	VAR_Process_Int_OK		S_SW4_No_Of_Operations
DATA_DOL_A_V1_0.VAR_Start_Int_OK	VAR_Start_Int_OK		S_SW5_Restart_PV
DATA_DOL_A_V1_0.VAR_Auto_Run_Req	VAR_Auto_Run_Req		S_SW6_Current_PV
DATA_DOL_A_V1_0.F.I_MAN_START_PB	F_I_MAN_START_PB		S_SW7_Running_Hours
DATA_DOL_A_V1_0.F.I_MAN_STOP_PB	F_I_MAN_STOP_PB		S_SW1_D_Fault_OverCurrent
DATA_DOL_A_V1_0.S.CW1_Marshal	S_CW1_Marshal		S_SW1_A_Fault_Start
DATA_DOL_A_V1_0.S.CW2_Fail_To_Start_SP	S_CW2_Fail_To_Start_SP		S_SW1_9_Fault_Stop
DATA_DOL_A_V1_0.S.CW3_Fail_To_Stop_SP	S_CW3_Fail_To_Stop_SP		S_CW1_0_Auto_Manual
DATA_DOL_A_V1_0.S.CW4_Running_SP	S_CW4_Running_SP		S_CW1_1_Desk_Field
DATA_DOL_A_V1_0.S.CW5_Stopping_SP	S_CW5_Stopping_SP		S_CW1_2_Maintenance
DATA_DOL_A_V1_0.S.CW6_Simulate_SP	S_CW6_Simulate_SP		S_CW1_3_Simulation
DATA_DOL_A_V1_0.S.CW7_Restart_SP	S_CW7_Restart_SP		S_SW1_1_Ready_To_Start
DATA_DOL_A_V1_0.S.CW8_MaxCurrent_SP	S_CW8_MaxCurrent_SP		S_SW1_2_Stopped
DATA_DOL_A_V1_0.S.CW9_MinIn_Current_SP	S_CW9_MinIn_Current_SP		S_SW1_3_Starting_RQ
DATA_DOL_A_V1_0.S.CW10_MaxIn_Current_SP	S_CW10_MaxIn_Current_SP		S_SW1_4_Running
DATA_DOL_A_V1_0.S.CW11_MinOut_Current_SP	S_CW11_MinOut_Current_SP		S_SW1_5_Stopping_RQ
DATA_DOL_A_V1_0.S.CW12_MaxOut_Current_SP	S_CW12_MaxOut_Current_SP		
DATA_DOL_A_V1_0.F.AI_Current	F_AI_Current_PV		

Template Graphic: Advanced Home Screen	Control	Descriptions
	Auto	Auto control by PLC
	Manual	Manual control by operator, desk and field mode
	Desk	Manual control from the SCADA faceplate
	Field	Manual control from the field start/stop station
	On	Manual start the DOL
	Off	Manual stop the DOL
	Reset	Reset the DOL after a fault has been repaired
	Healthy ###	Healthy to start, ### - Restart delay countdown in seconds
	Process Interlock	DOL process ok to run interlock ready
	Start Interlock	DOL ok to start interlock ready
	Start Fault	DOL failed to start fault
	Stop Fault	DOL failed to stop fault
	Isolator Fault	DOL isolator off fault
	MCC Fault	DOL MCC tripped fault
	Safety Fault	DOL Safety off fault
	Overcurrent	DOL overcurrent fault

Template Graphic: Advanced Setup Screen	Control	Descriptions
	Simulate	Enable simulation of DOL, disables the physical output
	Feedback Time	Time for simulation to simulate run feedback signal
	Maintenance	Enable maintenance mode (Disables all interlocks)
	Number of operations	Total number of DOL start operations
	Running Hours	Total number of DOL running hours
	Reset	Reset the total counter of the current service
	Start Signal Delay (SP)	Delay the PLC to accept the DOL is running
	Max Start (SP)	Time for running signal is on after start signal is on
	Last Start (PV)	Last time for running signal was on after start signal was on
	Stop Signal Delay (SP)	Delay the PLC to accept the DOL is stopped
	Max Stop (SP)	Time running signal is off after start signal is off
	Last Stop (PV)	Last time running signal was off after start signal was off
	Current Hi Limit	Sets the value of the maximum current for the DOL to trip at
	Current Value	Actual current the DOL is running at
	MinIn, MaxIn	Raw current input values for scaling
	MinOut, MaxOut	Engineering current values for scaling
	Restart (SP)	Restart prevention time of DOL
	Restart (PV)	Remaining time before DOL restart

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Current	Display the current of the DOL in the trend, with minimum, maximum and average values.
	Alarm Limit	Display the current Setpoint of alarming on high current of the DOL in the trend, with minimum, maximum and average values.

Typical Electrical Wiring Schematic Diagram of an Advanced DOL

